



GNC test

V2.2

Simple sensor fusion

motion.csv contains IMU readings.

The imu axes are oriented as follows:

- 0: forward
- 1: right
- 2: down

Create a python script to compute the estimated attitude and display the result as euler angles.

Your answer must include:

- the python code used to compute the attitude (.py files)
- a technical document that explains your algorithm with text, mathematical formulas, figures or any other information that will help understand your solution (.pdf)

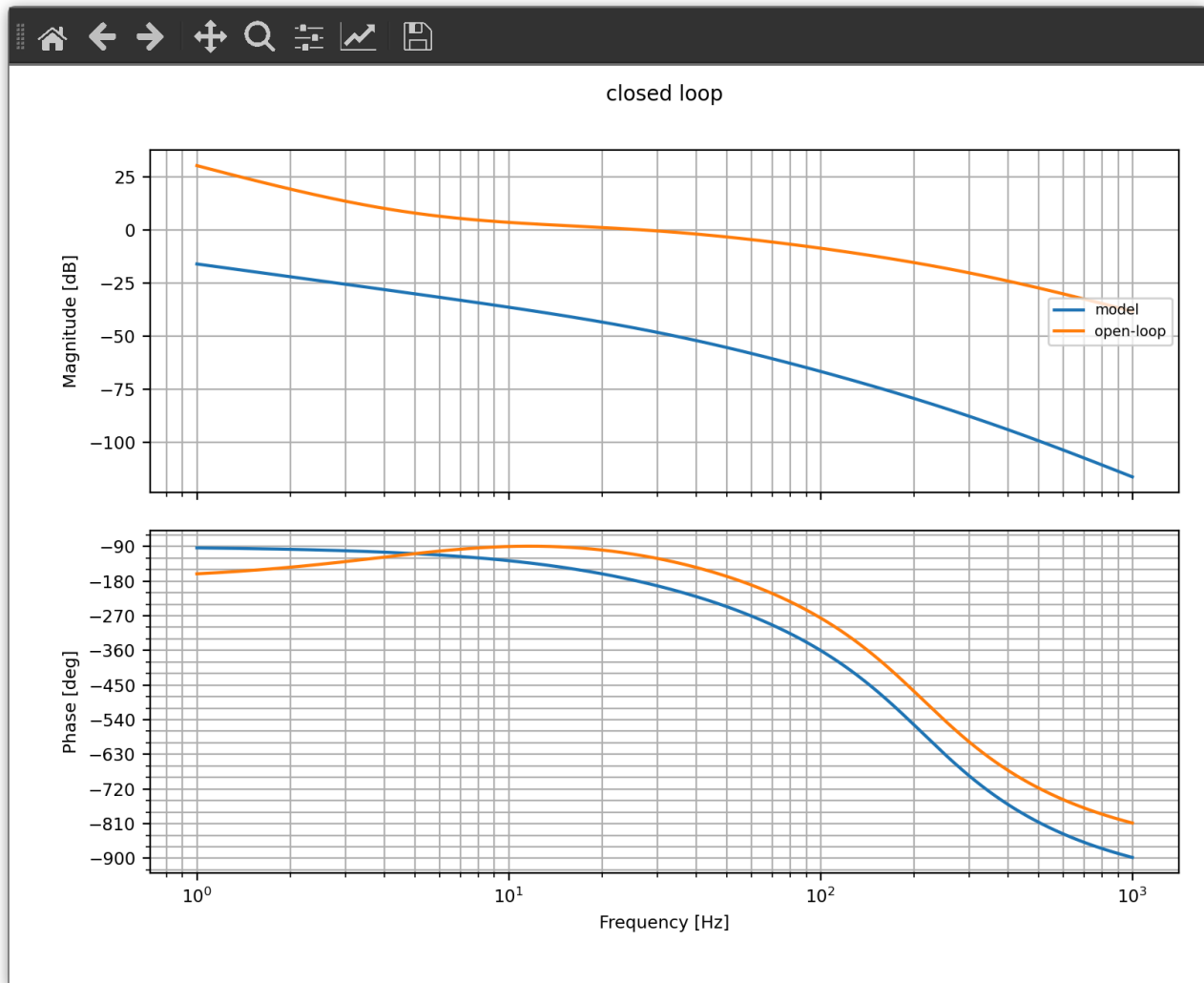


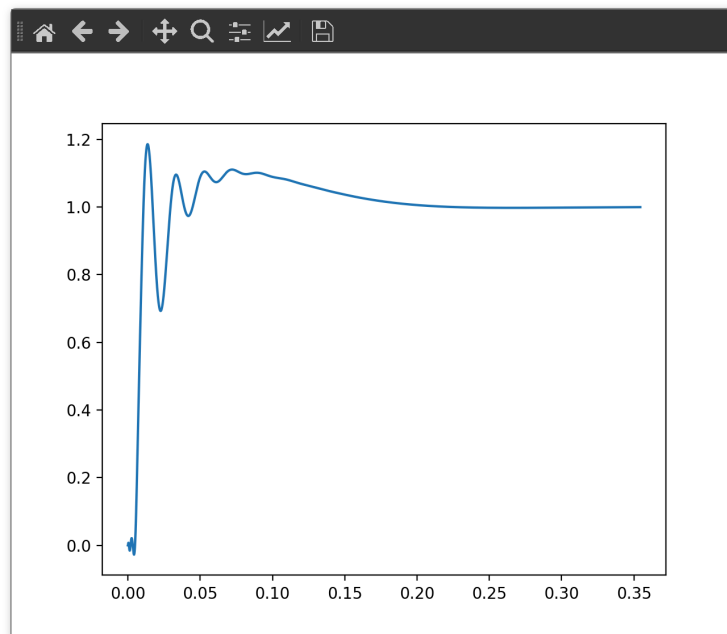
Fixing a control loop

Here is the bode plot representing the transfer function of a model and the open-loop (corrector and model) and the step response of the closed-loop system.

The corrector structure is a standard PID controller:

$$C(s) = K(1 + 1/(T_i * s) + T_d * s)$$



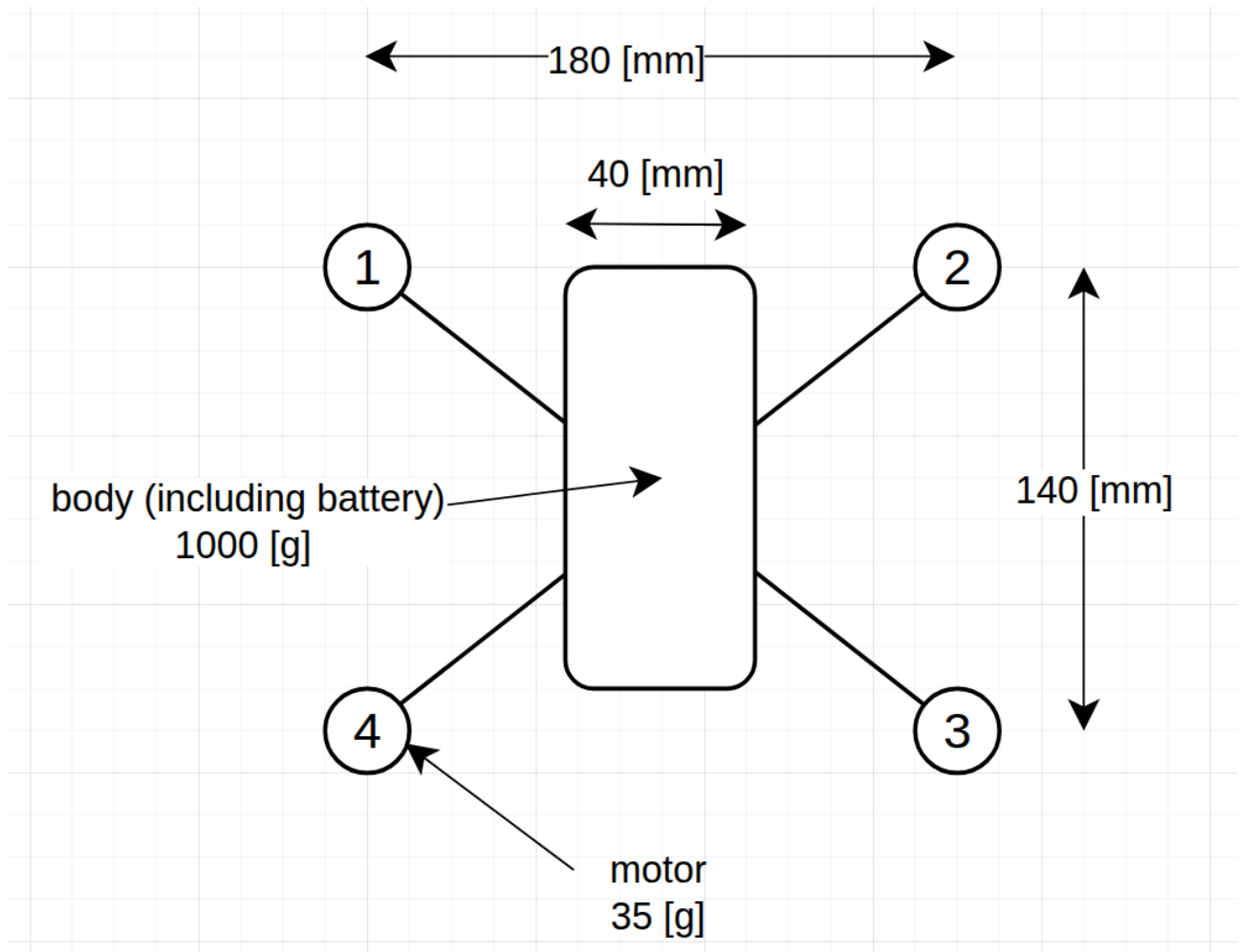


Analyze the current performances of the control loop and propose some improvements. Your answer must include a technical document with your analysis (.pdf file) and any python code you may have used (.py file).



Building a quadcopter model

The following graph gives an overview of a quadcopter drone:



Motors 1&3 are rotating clockwise, motors 2&4 are rotating counter clockwise.

Motor.csv contains some recording of the motor response to RPM steps.

Thrust.csv contains thrust measurements at different rotation speeds of the propeller.

1. What is the expected hovering RPM?
2. Use the following elements to build a linear MISO model of the roll and pitch angular rate loops as a function of RPMs (you may have to make some assumptions in case some information is missing or incorrect).
3. Linearize the MISO models around hovering point,



4. Decouple this system to get 2 SISO systems. What are the poles of the obtained transfer functions?
5. What are the poles and control margins of both transfer functions?
6. Design a regulator for both rate transfer functions.
7. How can you make sure that the proposed tuning is robust to the variations of the drone parameters? or unmodelled dynamics?

Your answer must include:

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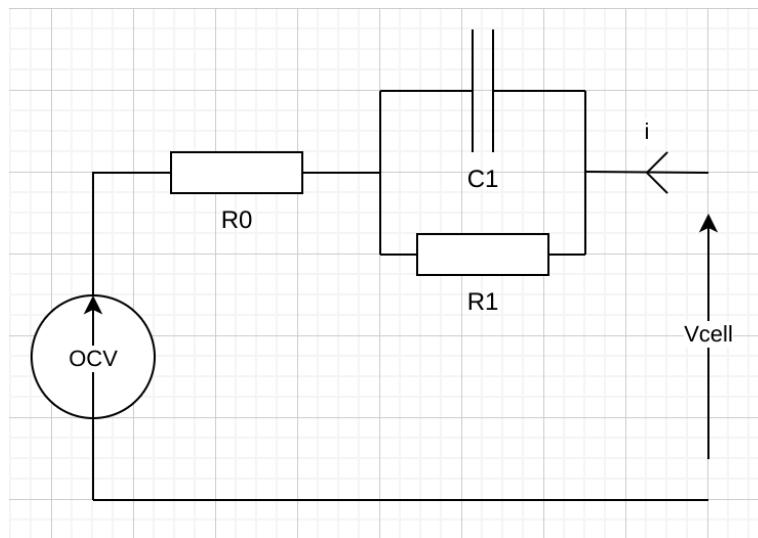
Battery Management System

battery_discharge.csv contains the result of a discharge test.

What is the battery capacity in [mAh]?

What is the battery topology?

We decide to use a first order equivalent circuit model for each battery cell.



Give an estimated value for R_0 .

Assuming $R_1 = R_0/2$, give an estimated value for C_1 .

OCV stands for open cell voltage. SoC stands for state of charge, 100% means the battery is full and 0% means the battery is empty.

Plot the function $OCV(\text{SoC})$.

Apply this function to the data in battery_flight.csv

Be careful, this is a battery with a different topology (4S1P) and capacity, but it is using the same chemistry so the $OCV(\text{SoC})$ function is the same.

What is the max cell voltage? What is the battery capacity?

Create a battery SoC estimator using an EKF that uses the current measurement for propagation of the model and the cell voltage as observation of the model.

Use the data provided in battery_flight to validate your algorithm.

Your answer must include:

- the python code used to compute the attitude (.py files)
- a technical document that explains your algorithm with text, mathematical formulas, figures or any other information that will help understand your solution (.pdf)